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Calculate atan(phase)

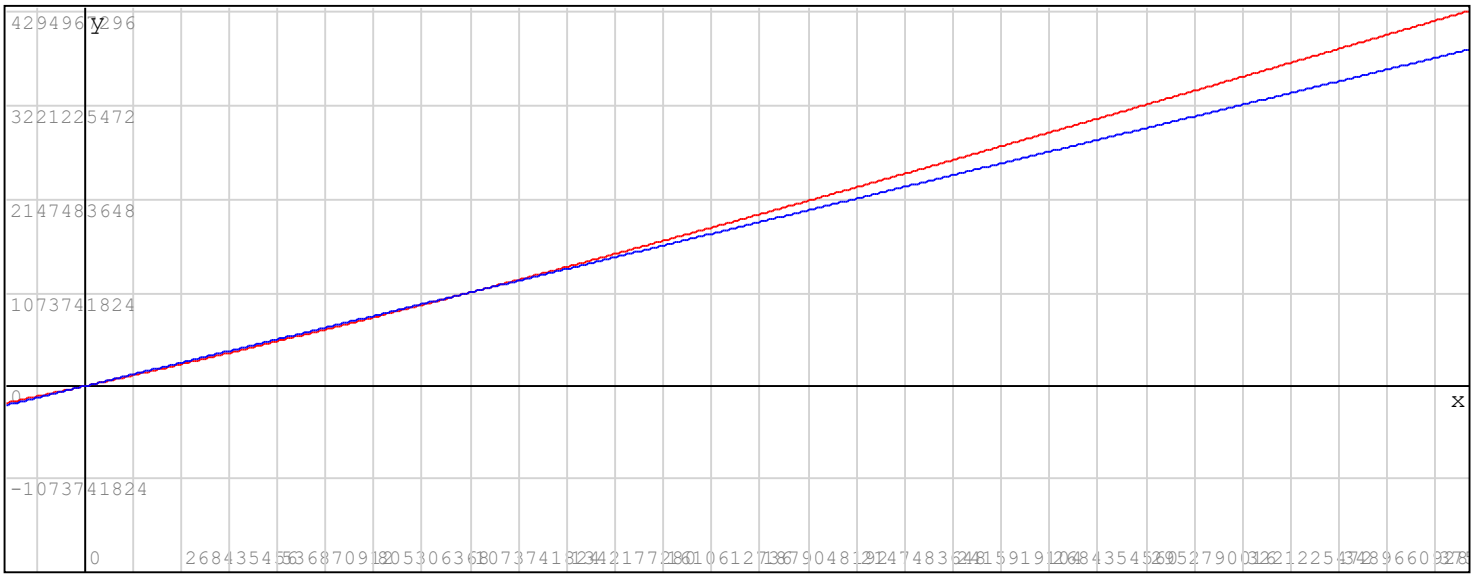
$j := \sqrt{-1}$ $S(phase) := \cos\left(\frac{phase}{2^{32}}\right)\cdot 2^{31} + j\cdot\sin\left(\frac{phase}{2^{32}}\right)\cdot 2^{31}$ $N := 0$ $ScaleCorr := 1.1549999999$

Atan2 approxymation

$ScaleCorr := 1.15533654773$

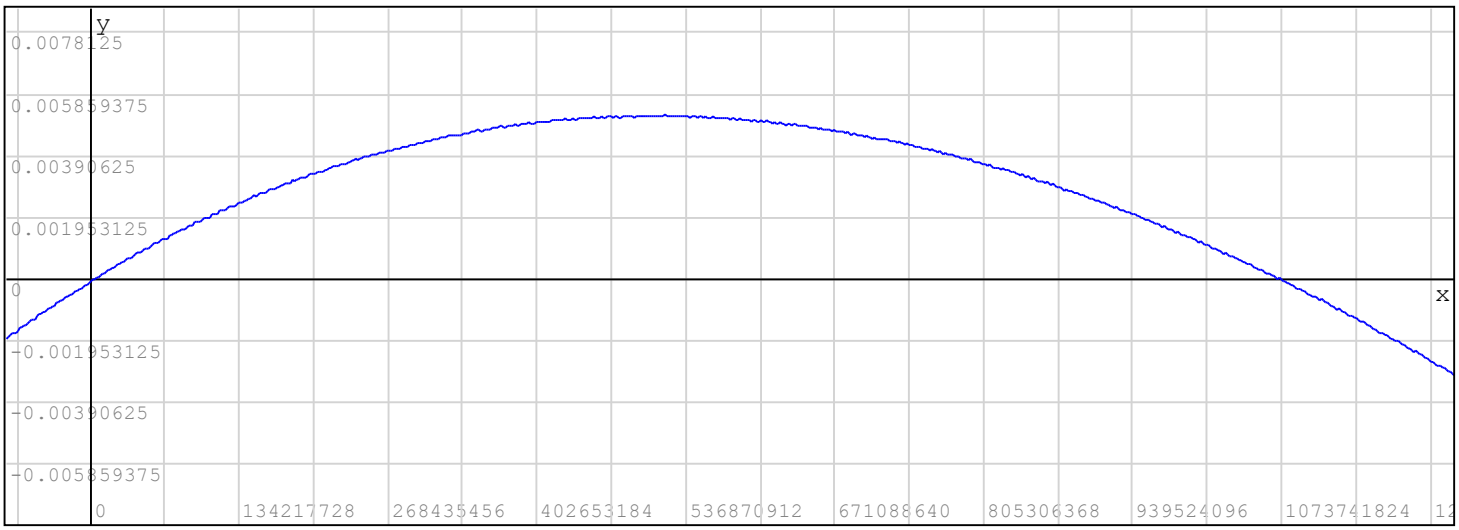
$$Phase(phase) := \left(2^{31} - \mathrm{trunc}\left(\frac{\mathrm{trunc}(\mathrm{Re}(S(phase)))}{\mathrm{trunc}\left(\frac{\mathrm{Im}(S(phase))}{2^{16-N}}\right) + \left(2^{16+N} - 0\right)}\right)\cdot 2^{16+N}\right)\cdot \mathrm{trunc}\left(\frac{\pi}{4}\cdot ScaleCorr\cdot 2^{32}\right)\cdot\frac{2^{2}}{2^{32}}$$

$Diff(phase) := phase - Phase(phase)$ $\frac{\pi}{4}\cdot ScaleCorr\cdot 2^{32} = 3.8972498999833\cdot 10^{9}$




$\begin{cases} x \\ Phase(x) \end{cases}$

$\frac{2^{32}}{4} = 1073741824$ $\frac{Diff\left(\frac{2^{32}}{4}\right)}{2^{32}} = 0$ $\frac{Diff\left(\frac{2^{32}}{2}\right)}{2^{32}} = -0.0301$




$\frac{Diff(x)}{2^{32}}$

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$Diff(x)$

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